

Theoretical context:

I have developed a line of zebrafish with a knockout of the SETBP1 gene. The aim of my project is to leverage this zebrafish line as a biomedical model for SETBP1 Haploinsufficiency disorder, a disorder caused by a SETBP1 knockout. My research questions are “what are the physiological differences between SETBP1 knockout and wildtype zebrafish? And what downstream genes are differentially regulated in the absence of SETBP1 in zebrafish?”

Relevant attributes and what to measure:

SETBP1 in humans is known to have a role in muscle development and function. A measurable way to look at that same role in zebrafish is to measure their critical swimming capacity (Ucrit). This is relevant because zebrafish with impaired muscle function have been shown to have a lower capacity for swimming.

SETBP1 in humans is a known transcription factor, regulating mitochondrial function genes (i.e. MT-RNR1). A measurable way to assess if SETBP1 is regulating these same genes in zebrafish is to complete transcriptomics, and measure differential expression between wildtype and SETBP1 knockout fish.

Ucrit –scale, cutoff level, and statistical analysis:

From a measurement theory perspective, Ucrit can be measured on a ratio scale, because there is a meaningful zero corresponding to the absence of sustained swimming when water flow is present, and equal differences in Ucrit represent equal absolute differences in swimming performance. Ucrit is a continuous physiological measure, so linear models are appropriate, and absolute differences in Ucrit retain biological meaning.

In the context of a disease model, I expect that there will be changes in swim capacity that may result from altered muscle structure or reduced physiological reserve, but I do not expect to see a complete inability to swim. Because of this, I want my threshold for significance to capture smaller changes in swim capacity between groups.

Because Ucrit is strongly influenced by body size, my primary analysis will use a linear mixed-effects model with genotype as a fixed effect and body length included as a covariate. Additional sources of variation, such as clutch or experimental day, will be included as random effects to account for non-independence and batch-level variability. This modeling approach is appropriate for ratio-scale data and allows me to estimate genotype-associated differences in swimming performance while accounting for known biological and experimental structure. I will report model-estimated differences in Ucrit in

absolute units, and I will also express these differences as percent change relative to wildtype to help interpret effect size relative to baseline swimming capacity.

My plan is to evaluate whether an observed difference wildtype and SETBP1 knockout fish is biologically important based on effect magnitude. Given the expected inter-individual variability in U_{crit} and the fact that muscle function is likely impaired but not eliminated in this disease model, I do not expect large differences between groups. I also expect very small changes (~5%) to be within the range of normal biological variation. I think that differences on the order of 10% or greater would suggest a biologically meaningful reduction in swim capacity. I will check this threshold by comparing the estimated genotype effect to the coefficient of variation observed in wildtype fish, if the effect size is similar to or larger than typical wildtype variability, this would support its biological relevance.

Transcriptomics – scale, cutoff level, and statistical analysis:

I would use a ratio scale to measure the differences in transcript levels between wildtype and SETBP1 knockout fish because it is possible for there to be no transcript, which is a meaningful zero.

I would log transform my measurements for transcript levels, as some genes are expressed many more fold times than other genes are. While some genes will be expressed more in both wildtype and SETBP1 knockout fish, what I care about seeing in my data is differences between wildtype and SETBP1 knockouts, so it is important to work on a scale that accentuates this. Because transcriptomic data is often very right skewed, the log-2 scale will work well.

For statistical analysis, I will use DESeq2. This models the raw counts using a negative binomial distribution. It produces log2 fold changes which is consistent with my choices from above. Log2 fold changes can be used to measure effect size. I am using a very small sample size for transcriptomics so I would use a conservative threshold for significance. I would want to see a large change, which I can set in DESeq2 (i.e. absolute log2 fold change set as ≥ 1).